A Deep Dive into Energy Optimization Through Exploring the Potential of Big Data Tools to Revolutionize Consumption Patterns

Pola Alaa Tobia, Abd Elrahman Ahmed Ibrahim, Youssef Ashraf ElNaggar, Ahmed Mohamed Abd Elazim, Abd Elrahman Salah, Ahmed Waeel Elmorshdy, Salma Khaled Hegazy, Khaled Mohamed Fouad

Abstract— Egypt faces a big energy problem due to increasing demand from cities and industries. Current solutions like random power cuts aren't working well because there's not enough data to make them effective. To tackle this, we're using data from England, which has a similar situation, to understand energy use better. This dataset shows when and how people use electricity, considering factors like where they live and the weather. We're using Hadoop and Apache Spark to handle this data effectively. We've identified risks like data quality issues and external factors, and we've planned steps to deal with them. Our project timeline outlines tasks from start to finish, aiming to develop ways to predict and manage Egypt's electricity demand more efficiently. Ultimately, we hope to improve energy sustainability and overcome supply problems.

I. Introduction

Egypt has faced a big problem for a long time as many people are using too much energy, and it's causing issues. With more people living in cities and more factories running, the demand for electricity keeps going up. One of the solutions that Egypt has already taken, is cutting the electricity off for around 2 hours a day in each region. Still, it’s not effective because they are cutting the electricity off with no information and analysis of this region.

The project aims to predict electricity consumption in Egypt depending on a reliable dataset. As a dataset from Egypt is not reliable, we sought a comparable dataset from a country with similar consumption patterns, landing on England. Both countries share similarities in population density and urbanization, as well as diverse socio-economic demographics. Although there are distinct differences between the two countries, England's dataset, arising from a comprehensive smart meter rollout initiative, offers valuable insights into consumer behavior and energy usage that could approximate or inform trends in Egypt.

II. Related Work

The smart meters are advanced meters which can record power consumption in real time and transmit the data back to the utility for monitoring and billing purposes. These gadgets form an essential part of contemporary energy management systems, hence enabling minute analysis and optimization of energy utilization.

The vast volumes of data generated by the smart meters can be used to identify energy saving techniques and optimize energy management. This method, which relies on data collected rather than intuition or guesswork, will enhance better decision-making both for consumers and energy providers as well as development of smart cities and sustainable energy practices. Granular insights about consumption patterns offered by smart meters help in identifying inefficiencies thereby promoting behaviors that save energy. For utilities, it helps them undertake demand forecasting, grid management and also cut down their operational costs through real-time data available to them all the time. For consumers, it enables making informed decisions related to their energy usage thus minimizing their electric bills as well as environmental pollution resulting from excessive use of electricity. Smart meters are critical elements in building a smart city with sustainability into its design incorporating renewable sources whilst contributing thus contributing towards global efforts against climate change ensuring reliable power supply systems.

In the subsequent sections, we'll explore papers investigating the relationship between smart meter data and energy consumption in residential contexts. We'll also delve into studies on weather's impact on energy usage dynamics. Additionally, we'll discuss research on smart cities and their implications for energy management.

A study conducted by researchers about using historical weather and consumption data to develop predictive models. They employed machine learning techniques such as regression analysis to quantify the relationship between weather variables and energy use. Their findings indicated a strong correlation between temperature fluctuations and electricity demand, suggesting that weather-based forecasting could enhance the accuracy of consumption predictions. However, the study faced challenges related to the variability of weather data and its integration with consumption records [3].

Furthermore, there is a study that analyzed the residential electricity demand in London by integrating smart meter data with demographic information. The problem they addressed was the variability in energy consumption patterns across different households. They employed clustering techniques to identify distinct consumption profiles and used regression models to examine the influence of demographic factors on energy use. Their methodology included data pre-processing, feature extraction, and the application of k-means clustering. The results highlighted significant variations in consumption patterns, which could inform targeted energy-saving measures. However, the study was limited by the granularity of demographic data and the short time span of the smart meter data used [5].

Another paper presents a proof-of-concept prototype for a big data management system tailored for handling large-scale energy data, particularly focusing on electricity usage in industrial complexes in Korea. It details the architecture of the system, which includes cluster systems and distributed file systems like Hadoosp, along with the integration of HBase for distributed database management. The system also incorporates web-based data access APIs for querying and retrieving energy usage data. Additionally, the paper

introduces a novel energy consumption prediction model based on penalized linear regression and implemented using map/reduce algorithms. Experimental results demonstrate the effectiveness of the prediction model, showcasing its performance in accurately predicting energy consumption patterns. The paper concludes with plans for future enhancements, including scaling up the system, incorporating additional features for prediction, and improving the user interface [6].

IV. Methodology

Design of the Data Flow Architecture

Initially, the raw data undergoes storage in HDFS within the Hadoop cluster. HDFS, designed meticulously to accommodate extensive datasets, operates by distributing data across multiple nodes within the cluster. This distributed storage approach not only ensures efficient space utilization but also facilitates fault tolerance through data replication. Each data block, typically sized at 64 MB, though configurable, is replicated across distinct nodes to enhance reliability. This replication process, with a default factor of 3, ensures that data remains accessible even in the event of node failures, thereby bolstering the system's resilience.

The distributed nature of HDFS is integral to its parallel processing capabilities, enabling seamless computation of vast datasets across the cluster. The reason is that HDFS enables simultaneous data analysis by breaking down data into manageable chunks and distributing them across nodes. Additionally, its scalability ensures that the system can handle bigger datasets without compromising on performance. HDFS's fault tolerance capabilities and scalability make it a good option for big data enterprise workloads.

Enter Apache Spark, a distributed computing platform aimed at overcoming the limitations of traditional MapReduce processing. Its key innovation is to support in-memory processing which stores intermediate computations in memory rather than writing them to disk after each operation. This approach eliminates much of the overhead associated with I/O operations and improves performance significantly, especially when one has iterative algorithms or interactive analysis tasks.

I. Data Acquisition

Data acquisition revolves around gathering real-time energy consumption data from smart meters installed in households and businesses. These smart meters record electricity consumption readings at regular intervals, capturing data on a half-hourly, hourly, and daily basis. This comprehensive dataset provides detailed insights into energy usage patterns over varying time intervals.

II. Data Storage

The collected data is stored in a distributed file system, such as Hadoop Distributed File System (HDFS), for scalability and fault tolerance. Robust security measures, including encryption and access controls, are implemented to protect sensitive information stored in the system. This distributed storage architecture enhances reliability and flexibility while facilitating parallel processing of large datasets.

Data Preprocessing:

This section outlines the approach that was suggested in order to identify trends in the time series of big data related to power use. Specifically, this approach uses the k-means contained in the Apache Spark Machine Learning Library to produce clusters that are then studied to determine patterns of power usage.

We made some EDA to understand our dataset and working on it correctly

First we made sure that the flow of every numerical column is normal

In addition to in-memory processing abilities, Spark directly interfaces with HDFS providing access to data stored in the Hadoop cluster making it possible for Spark to take advantage of distributed storage features offered by HDFS hence effective handling of data dynamics within an organization without necessitating expensive duplication or transferring. Furthermore, Spark's ability to cache intermediate results in memory further enhances its performance by minimizing redundant computations and facilitating faster data access.

To harness the power of Spark for data processing and analysis, Jupyter Notebooks will be used as our development environment. Jupyter provides an interactive platform that seamlessly integrates Spark code with explanatory text, visualizations, and other analytical tools. This interactive environment fosters rapid prototyping, experimentation, and collaboration on Spark-based projects, empowering data scientists and analysts to explore complex datasets and derive valuable insights with ease.

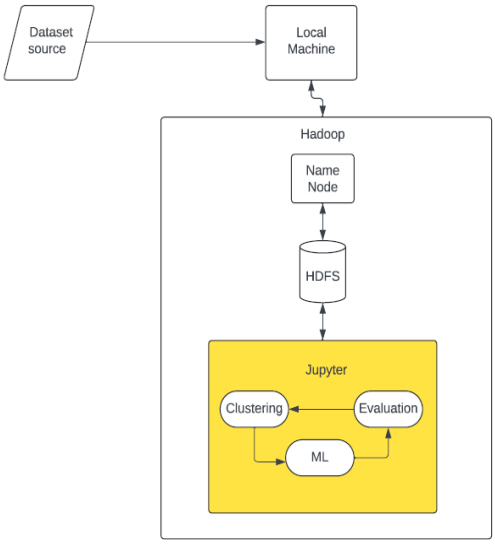
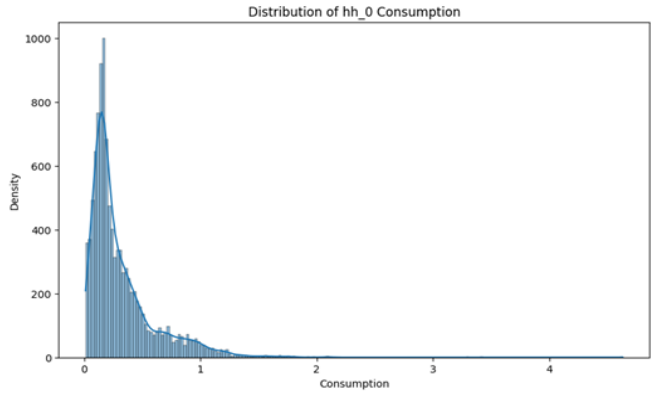
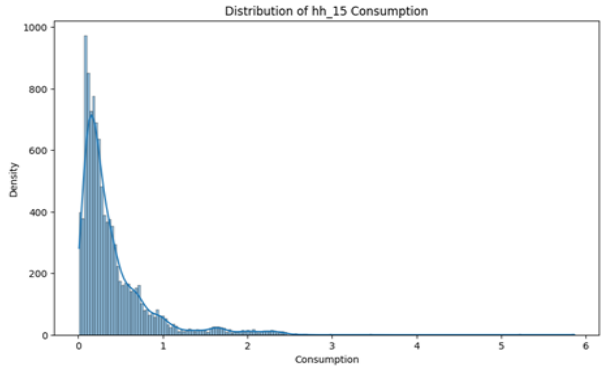


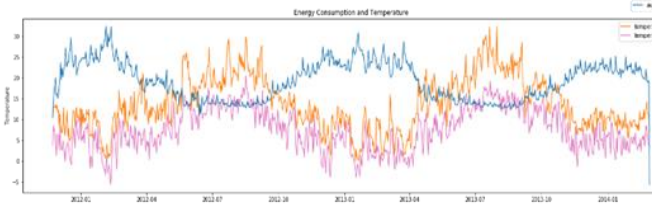
Fig.1: Design of the Data Flow Architecture



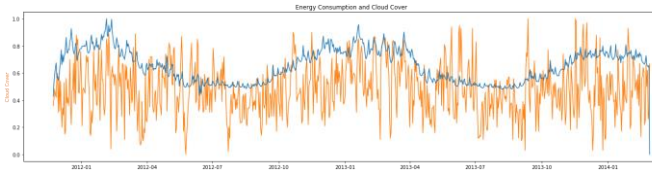
We done the same to the 15th half hour column and all half hours columns, we found that all half hours has the same distribution.



One of the most features that affect the consumption of energy is temperature, we found that all over the years the temperature and energy have inverse relationship,



The cloud cover value and Humidity seem to be following the same pattern as the energy consumption.



Preprocessing the data is the initial step. This phase's goal is to clean up and modify the original dataset in order to produce an RDD variable that Spark can distribute over a cluster and process. Several CSV files were processed to get the original dataset. These files included records of power usage statistics from around 5000 public buildings presented as a time

series. Data were extracted from the smart meters installed in the buildings.

The smart metres that were put in the buildings provided the data. Between 2011 and 2014, data on power use was gathered by smart metres every thirty minutes. The building name, the date and time (split into 0 to 47 half hours Fetures), and the energy consumption information at that particular moment make up the initial RDD variable. In order to generate accurate learning models for the subsequent stages, our program pre-processed the rows containing missing information and gathered consumption data during the data purification step.

This cleaning phase will be discussed extensively in the results section. Before creating the model, it is necessary to perform a transformation in the original dataset by grouping the energy consumption series into rows of around 3.5 million records corresponding to readings half hourly each day. As each day contains the description of the smart meter id, days, and the reading of each half hour the original dataset has a total of 3.5 million records.

Obtaining the Optimal Number of Clusters:

The second phase of the methodology consists in obtaining the optimal number of clusters for the dataset by analyzing and interpreting various CVIs. However, some CVIs have limitations to be applied to large datasets due to the computational costs of quadratic complexity. This cost could take much longer to apply than the clustering algorithm used in this study. For this reason, we have applied big data clustering validity indices (BD-CVIs). In this paper we analyze the results of four BD-CVIs. Three of them are based on traditional CVIs—the BD-Silhouette, BD-Dunn and Davies-Bouldin indices—and the other is based on the Within Set Sum of Square Error (WSSSE) index offered by the MLlib. These BD-CVIs will be defined below.

Majority Voting Methodology:

The aim is to apply this group of indices to the complete set of data so that we can validate them and also obtain the optimal number of clusters K, which will be used as an input parameter of the parallelized k-means algorithm. In this sense, this work proposes a methodology of majority voting, which combines the results obtained of the application of the four indices above as a single result. The voting strategy is now explained.

The application of each of the indices separately to the complete dataset generates a graph that will show maxima or minima indicating the optimal number of clusters according to each case. Therefore, each of the graphs will have a first best value, a second best value, a third best value, and so on. The voting system will evaluate the best results of all indices so that we extract as a result of the optimal k number of clusters to group our dataset. There is a favourable and ideal situation: that all indices coincide with the best value or that most (i.e., at least three) coincide. In this case, we will take this value as the optimum k.

However, a second situation may arise: there are no coincidences or these are the minority (that is, fewer than three). When this case occurs, in addition to the first best values, the second best results of the four indices will be also

considered. If a majority is not reached, we will study the third best results, and so on until we find a majority that matches. The selected k will then be the one that is repeated most times until the majority is found.

Table 3 provides an example of how this system is being used. The second circumstance arises in this instance: the application of the BD-Silhouette (four clusters) and WSSSE (seven clusters) indices differs from the first two since only the BD-Dunn (six clusters) and Davies-Bouldin (six clusters) indices provide the same result (i.e., the minority of the first best results).

In order to determine the ideal number of clusters at this stage, we must examine the second-best findings. The BD-Silhouette index, which provides six clusters as the second-best result, corresponds with the BD-Dunn and Davies-Bouldin indices when we examine the second-best values. Consequently, after looking at the first- and second-best matches, we will have discovered a majority, or at least three matches.

Table 3. Majority voting methodology.

Values	BD-Silhouette	BD-Dunn	Davies-Bouldin	WSSSE
First	4	6	6	7
Second	6	8	9	15
Third	9	13	15	21

Applying Machine Learning Using of MLlib

Once the optimal number of clusters k for the dataset has been obtained, the clustering algorithm can be applied. The algorithm used for discovering patterns from the dataset is the k-means. This algorithm is a parallelized version of the k-means included in the MLlib of Apache Spark. This clustering algorithm is based on the classic k-means algorithm and has been developed to extract patterns in parallel and distributed systems. Figure 2 shows how a run of the k-means works. First, the RDD object containing the complete dataset is distributed in several slave nodes for the execution of k-means, obtaining initial centroids n. Second, the Apache Spark engine shuffles the resulting n centroids for each run. Finally, the k-means algorithm computes the WSSSE index in each partition for each centroid, returning the one that minimizes the WSSSE as the best. It is worth remembering that there are as many centroids as there are concurrent executions.

Centroids #1 The last phase corresponds to interpreting and evaluating the results obtained after the application of k-means with the optimal number of clusters to the dataset. We will obtain and analyze different types of results to obtain electric consumption patterns in big data time series. We will obtain the distribution of instances in each cluster and the centroids of the daily electricity consumption clusters. Although clustering is considered an unsupervised learning technique, a clustering validity analysis has been carried out in this study, using features of the instances such as a type of day, season, or building as labels. Clustering results are merged with the features that each instance could have. Table 4 shows an example of the data that will be analyzed. Each row represents an instance of the dataset, i.e., the electricity consumption of a day, and the column cluster indicates the cluster assigned to that consumption. Also, each instance includes the features to be analyzed such as the building in which electricity was consumed, the season of the year, and the day of the week or non-working day.

Table 3. Example of a dataset along with assigned cluster and features.

ID	Cluster	Building	Season	Day
1	1	Build_1	Summer	Day off
2	1	Build_1	Winter	Day off
3	2	Build_20	Summer	Thursday
4	1	Build_42	Summer	Friday
5	3	Build_1	Autumn	Monday

With this information, we can see how the clusters are built regarding the features. Following our example, we can observe how the buildings are distributed by clusters, check out in which cluster there are more days off, or determine if the clusters are influenced by the season of the year.

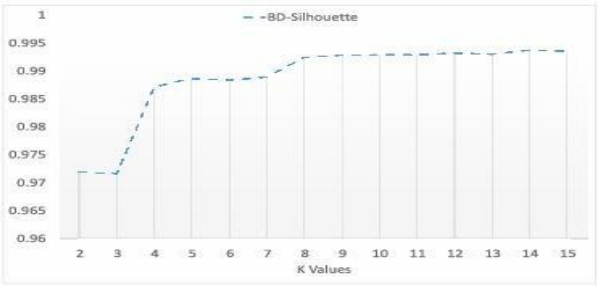
The evaluation of our linear regression model, used to analyze relationships between variables and make numerical predictions, demonstrated robust performance. We assessed the model using Root Mean Squared Error (RMSE) on both the training and test datasets. The results showed a Train RMSE of 0.012 and a Test RMSE of 0.015. These low RMSE values indicate the model's high accuracy in predicting outcomes on both the training and test datasets, suggesting that the linear regression model performs well, providing accurate predictions with minimal error.

Based on this reasoning, we will draw general conclusions using percentages of the distribution of buildings in the k clusters. We will also conduct a study of several synthetic big datasets. Starting from the base of the original set, we will multiply its original size with the objective of checking the efficiency in computing time of the proposed methodology.

V. Results

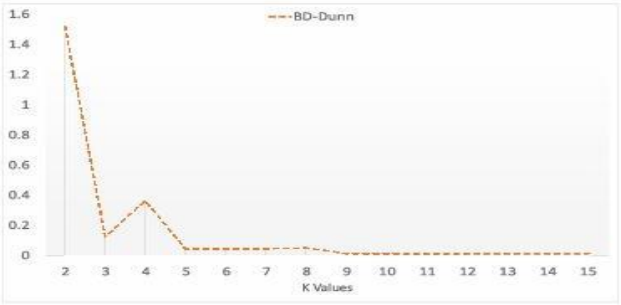
This methodology aims to characterize electricity consumption over time, providing useful insights for efficient energy resource management, and represents the initial step towards achieving the desired results of this paper. Determining the optimal number of clusters is essential, with this study using BD-CVIs, including BD-Silhouette, BD-Dunn, Davies-Bouldin, and WSSSE indices.

Fig.3: BD-Silhouette suggests 4 or 8 clusters as optimal.



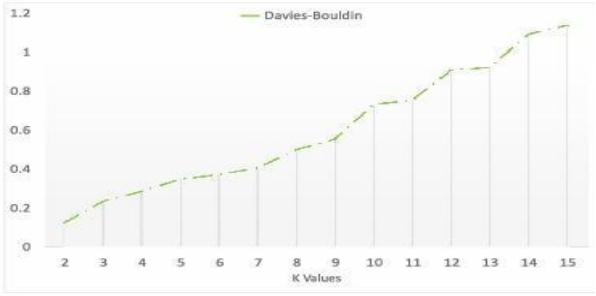
(a) BD-Silhouette.

Fig. 4: BD-Dunn suggests 4 or 8 clusters as optimal.



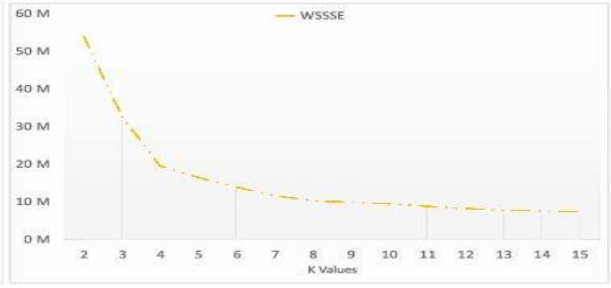
(b) BD-Dunn.

Fig.5:Davies-Bouldin does not offer clear results.



(c) Davies-Bouldin.

Figure. 6: WSSSE suggests 4 or 8 clusters as optimal.



(d) WSSSE.

Model Comparison and Analysis

In our analysis, we compared different clustering algorithms and evaluated the performance of linear regression and random forest models on the same four clusters. The objective was to understand the relationships between variables within the dataset and predict numerical outcomes based on input features.

We employed a linear regression model to examine the relationships between the dependent variable and multiple independent variables. Linear regression, a fundamental technique in statistical modeling, is valuable for understanding the strength and nature of these relationships. By utilizing this model, we aimed to predict numerical outcomes and gain a deeper understanding of the underlying patterns in the dataset.

Train Root Mean Squared Error (RMSE): 3.2591

Test Root Mean Squared Error (RMSE): 3.8378

We also utilized a Random Forest model to analyze the dataset. Random Forest, a robust ensemble learning technique, is particularly adept at handling complex datasets with high dimensionality. It constructs multiple decision trees during training and outputs the mean prediction of these trees. By leveraging this model, we aimed to enhance prediction accuracy and improve our understanding of the dataset's patterns through its ability to capture non-linear relationships and reduce overfitting.

Train Root Mean Squared Error (RMSE): 1.3545

Test Root Mean Squared Error (RMSE): 2.2657

A key challenge was addressing zero-value measurements due to smart meter errors, resolved by replacing them with the mean of the subsequent high values divided by the number of empty measurements. The four-cluster analysis identified one cluster with 72% of instances showing low consumption on non-working days, while the other clusters had high consumption with seasonal distinctions. The eight-cluster analysis revealed two main clusters with 39% and 32% of instances, linked to different buildings, seasons, and day types. The centroids highlighted clusters with high daylight consumption and others with consistent low consumption. Figure 2 details these findings: Cluster 1 shows the lowest, constant consumption throughout the day for all buildings except research centers 20 and 21, mainly on non-working days.

Clusters 2, 3, 4, and 6 from building 20 have the highest daylight consumption, with Clusters 2 and 6 covering all days, and Clusters 3 and 4 specific to working and non-working days, respectively. Cluster 5, from building 21, has low daylight consumption across all days, slightly more on non-working days. Cluster 7, excluding buildings 20, 21, and 42, shows low daylight consumption mainly on working days. Cluster 8, from buildings 11 and 21, indicates low daylight consumption, particularly on non-working days.

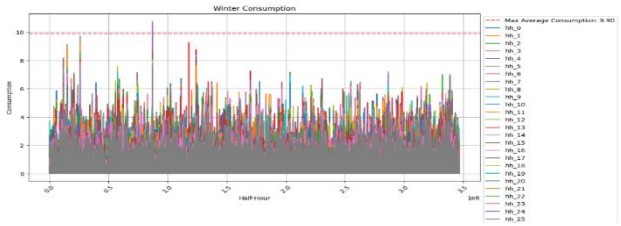
Fig.7

Cluster	Consumption	Days	Seasons
1	Low	Non-Working	Non-Summer
2	High, Diumal	Working	

3		High, Diumal	Working	Summer
4		High	Non-Working	
5		Low, Diumal		
6		High, Diumal		Non-winter
7		Low, Diumal	Working	
8		Low, Diumal		

The half-hourly energy consumption data from the hh.csv[9] dataframe was categorized into seasonal segments (winter, summer, fall, and spring), revealing notable patterns. Analysis indicates that the most significant energy consumption consistently occurs between 4 PM and 6 PM across all seasons, with winter exhibiting the highest overall consumption levels. See Figure 5 to which is the maximum electricity half hour consumption. Concurrently, insights from the weather_daily_darksky.csv[9] file highlight temperature patterns by season. The data shows that the average maximum temperature peaks around 13:51 in autumn, 14:36 in spring, 14:37 in summer, and 13:53 in winter. Conversely, average minimum temperatures are typically recorded at 7:39 in autumn, 5:39 in spring, 4:45 in summer, and 9:15 in winter. These timings coincide with periods when people are likely to use a significant amount of electricity for heating and cooling purposes, given the extremes in temperature. The alignment of high consumption periods with temperature extremes underscores the importance of understanding seasonal varifations in energy usage for effective energy management and planning.

Fig.8



From Figure 8, we know that the peak 2 hours at every time of every day specially winter season is from 4 : 6 pm.

VI. Conclusion

In conclusion, this project aimed to develop data-driven approaches to predict and manage electricity consumption patterns in Egypt more effectively. By leveraging a comprehensive dataset from England and employing big data technologies like Apache Hadoop and Apache Spark, we successfully implemented a robust data processing pipeline. The application of big data clustering validity indices (BD-CVIs) enabled us to determine the optimal number of clusters for the dataset. Subsequently, the parallelized k-means algorithm from Apache Spark's MLlib was applied,revealing distinct electricity consumption profiles influenced by factors such as building types, seasons, and day types. While this study focused on the England dataset, the methodology can be adapted to Egyptian data once a reliable source becomes available. The findings lay the groundwork for developing predictive models and optimizing energy management strategies tailored to Egypt's context. Overall, this project demonstrated the potential of harnessing big data and machine learning techniques in the domain of energy analytics. The insights gained represent a significant step towards understanding and managing electricity consumption patterns, contributing to sustainable and efficient energy resource management practices.

REFERENCES

[1] Pérez-Chacón, R., Luna-Romera, J. M., Troncoso, A., Martínez-Álvarez, F., & Riquelme, J. C. (2018). Big data analytics for discovering electricity consumption patterns in smart cities. Energies, 11(3), 683.

[2] Rahman, M. N., Esmailpour, A., & Zhao, J. (2016). Machine learning with big data an efficient electricity generation forecasting system. Big Data Research, 5, 9-15.

[3] Karim, B. A., Patel, R., & Seraphim, M. B. I. (2021). Weather based Electricity Consumption Prediction using Big Data Analytics Model. Int. J. Eng. Res. Technol, 10(4), 158-163.

[4] J.-M. D., "Smart meters in London Part 1: Description and first insights," Medium, Feb. 13, 2020. [Online]. Available: https://medium.com/@boitemailjeanmid/smart-meters-in-london-part1-description-and-first-insights-jean-michel-d-db97af2de71b

[5] M. Sun, I. Konstantelos, and G. Strbac, "Analysis of diversified residential demand in London using smart meter and demographic data," in 2016 IEEE Power and Energy Society General Meeting (PESGM), 2016, doi: 10.1109/pesgm.2016.7741076.

[6] Lee, W., On, B. W., Lee, I., & Choi, J. (2014, September). A big data management system for energy consumption prediction models. In Ninth International Conference on Digital Information Management (ICDIM 2014) (pp. 156-161). IEEE.

[7] R. Mohammad, "AMI Smart Meter Big Data Analytics for Time Series of Electricity Consumption," in Proc. 2018 17th IEEE Int. Conf. Trust, Security and Privacy in Computing and Communications/ 12th IEEE Int. Conf. Big Data Sci. and Eng. (TrustCom/BigDataSE), 2018, pp. 1875-1882, doi: 10.1109/TrustCom/BigDataSE.2018.00267.

[8] M. C. Pegalajar, L. G. B. Ruiz, M. P. Cuéllar, and R. Rueda, "Analysis and enhanced prediction of the Spanish Electricity Network through Big Data and Machine Learning techniques," Int. J. Approx. Reason., vol. 133, pp. 48–59, 2021, doi: 10.1016/j.ijar.2021.03.002.

[9] J. Midev, "Smart meters in London," Kaggle, available: https://www.kaggle.com/datasets/jeanmidev/smart-meters-in-london/data. [Accessed: 20/2/2024]